FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Main)-2021

TEST DATE: 23-12-2020

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A

2. D

Sol. For non-uniform circular motion the total acceleration will be the resultant of radial and tangential. So it need not be directed towards the centre but for the radial acceleration we need a variable force as the acceleration is always directed towards the center.

3. C

4. A

5. B

Sol. $A_n = \pi r^2$

$A_n = \pi \left(r_r n^2 \right)^2$

$A_n = \pi r_0^2 n^4$

So $\ln \frac{A_n}{A_1} = 4 \ln\ n$

6. C

Sol. $\eta = 1 - \frac{T_2}{T_1}$

$0.8 = 1 - \frac{T_2}{T_1}$

$\frac{T_2}{T_1} = 0.2$

$T_2 V_2^{\gamma-1} = T_1 V_1^{\gamma-1}$

$\left(\frac{V_2}{V_1}\right)^{\gamma-1} = \frac{T_1}{T_2} = 5$

$\frac{V_1}{V_2} = \left(\frac{1}{5}\right)^{5/2}$

7. D

Sol. $\omega' = \sqrt{\frac{K}{m} - \frac{b^2}{2m^2}}$ and $\omega' = \frac{2\pi}{T}$

8. B

Sol. Zero error = L.C. × division above surface line

Zero error = $10^3 \times 7 = +7 \times 10^{-3}\, cm$

9. D

Sol. $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \rightarrow$ differentiating twice w.r.t. time

$|a_i| = \frac{2m^3 u_0^2}{f}$

$u_0 \rightarrow$ volume of object

$m \rightarrow$ magnification

$f \rightarrow$ focal length.

10. A

Sol. $\tau = MB = C\theta$

$\Rightarrow NiAB = C\frac{\pi}{2} \Rightarrow C = \frac{2NiAB}{\pi} .....(1)$

When Q is passed,

$\int \tau\, dt = \int NiABdt$

$I\omega = N\,A\,B\,Q$

Also, maximum deflection happens when entire $\frac{1}{2} I\omega^2$ converts to $\frac{1}{2} C\theta^2_{\max}$

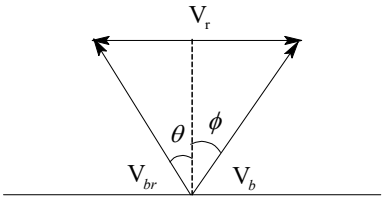
So $\theta_{\max} = \sqrt{\frac{I\omega^2}{C}} = \frac{1}{\sqrt{IC}} I\omega = \frac{NABQ}{\sqrt{IC}}$

11. B

Sol. Drift $X=\left(V_r - V_{br}\sin\theta\right)\dfrac{d}{V_{br}\cos\theta}$

For minimum drift if $V_r > V_{br}$

$\dfrac{dx}{d\theta}=0 \Rightarrow \sin\theta=\dfrac{V_{br}}{V_r}$



$\sin\theta=\dfrac{4}{5}\Rightarrow\theta=53^\circ\left(143^\circ \text{ with the flow direction}\right)$

Time $=\dfrac{36}{4\cos 53^\circ}=\dfrac{36}{4\left(3/5\right)}=15\sec$

$X_{\min}=\left[5-4\left(\dfrac{4}{5}\right)\right]15=\dfrac{9}{5}\times 15=27m$

12. B

13. D

14. A

15. B

Sol. First minima $=\dfrac{D\lambda}{a}$

First maxima $=\dfrac{3D\lambda}{2a}$

$\Rightarrow\dfrac{3D\lambda}{2a}=\dfrac{\left(540\right)D}{a}\Rightarrow\lambda=360nm$

16. B

Sol. $I=\dfrac{5}{4}MR^2$

$\dfrac{aI}{I}=\dfrac{aM}{M}+\dfrac{2dR}{R}=1\%+2\left(1.5\%\right)\Rightarrow 4\%$

17. C

Sol. $L_sR_s=i_gR_g$

$i_s=100A$

18. D

Sol. After passing through first shot $I = I_0 / 2$

$$\Rightarrow \frac{I_0}{4} = \frac{I_0}{2}\cos^2\phi$$

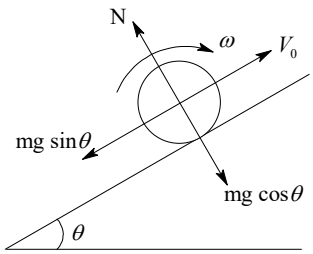
$$\Rightarrow \cos^2\phi = \frac{1}{2} \Rightarrow \phi = 45°$$

19. D

Sol. Since initial momentum is zero so final momentum is also use.
So momentum of hydrogen atom and photon will be equal.

20. A

Sol. Retarding force for linear velocity = mg sin θ. All the forces pass through the centre. There is no force to provide moment to reduce angular velocity. So angular velocity will remain constant up to the highest point. The sphere will stop going up at a point where translational velocity becomes zero.



$$\frac{m(V_0 - 0)}{t} = mg\sin\theta \text{ or } t = V_0 / g\sin\theta$$

SECTION – B

21. 3

Sol. $$\text{time} = \frac{\text{angle travelled}}{\text{angle speed}} = \frac{\pi/2}{2\pi/12} = 3\,\text{sec}$$

22. 4

SECTION – C

23. 00000.66

Sol. $I = \chi H$ and $\chi = \frac{C}{T}$ (according to curie law)

Magnetization (T)

$$\propto \frac{\text{Magnetic Induction (B)}}{\text{Absolute temperature (T)}}$$

$$\frac{I_2}{I_1} = \frac{B_2 T_1}{B_1 T_2}$$

24. 00001.20

Sol. $N\theta = 2n\pi$

$N \rightarrow$ side of regular polygon

$N \rightarrow$ number of rotation

$$\frac{N}{n} = \frac{360}{74} = \frac{180}{37}$$

Time taken to travel one side of a polygon $= \frac{180}{180} = \frac{1}{10}\sec$

Speed of particle $= 100 m/s$

$$\langle a \rangle = \frac{2v\sin\theta/2}{\Delta t} \simeq \frac{20\sin 37^\circ}{1/5}$$

$$\langle a \rangle = 60\, m/s^2$$

25. 00000.50

Sol. $$\phi = \left[\vec{E}.\hat{n}\right] A = \left[\frac{8\times 2 + (6\times 3) + (10\times 6)}{\sqrt{2^2+3^2+6^2}}\right]\frac{7}{47}$$

Chemistry

PART – II

SECTION – A

26. C

27. B

Sol. $I_2 + \text{dil.}OH^- \longrightarrow I^- + IO^- + H_2O$

28. D

29. C

Sol. $Mg_2C_3 + H_2O \longrightarrow Mg(OH)_2 + C_3H_4$

30. D

Sol. Inert pair effect

31. B

32. C

33. D

Sol. $T_c = \frac{8a}{27Rb}$ comes out to be more then $200°C$ option D is correct.

34. A

35. D

36. A

37. B

Sol. $C_xH_y + \left(\frac{2x + \frac{4}{2}}{2}\right)O_2 \longrightarrow x\,CO_2 + \frac{4}{2}H_2O$

Volume of $CO_2 = 100 - 60 = 40\,ml$

Molar ratio of $CO_2 : H_2O = 40 : 60 = 2 : 3$

38. B

Sol. $BO_3^{3-} + CH_3OH \xrightarrow{H^+} B(OMe)_3$

39. C

40. A

Sol. $\underset{(X)}{SO_3^{2-}} + 2H^+ \longrightarrow H_2O + \underset{(Y)}{SO_2}$

SO_2 is a colourless gas with irritating smell.

$$3SO_2 + K_2Cr_2O_7 + H_2SO_4 \longrightarrow \underset{\text{(Green)}}{Cr_2(SO_4)_3} + K_2SO_4 + H_2O$$

41. D

42. C

Sol. Because Cr(+VI) has d° configuration hence d-d transition will not possible.

43. B

44. D

45. A

SECTION – B

46. 3

Sol. $V \propto n$

$$\frac{\text{Anode}}{\text{Cathode}} = 1\text{mole}(C_2H_6) + 2\text{mole}(CO_2) / 1\text{mole}(H_2)$$

47. 6

Sol. Let we assume one corner of unit cell at origin so the corners of units cell will be at A (0, 0, 0) B ($8\sqrt{3}$, 0, 0), C($8\sqrt{3}$, 0, $8\sqrt{3}$), D (0, 0, $8\sqrt{3}$), w (0, $8\sqrt{3}$, $8\sqrt{3}$), x ($8\sqrt{3}$, $8\sqrt{3}$, $8\sqrt{3}$), y($8\sqrt{3}$, $8\sqrt{3}$, 0), z (0, $8\sqrt{3}$, 0), Now two nearest octahedral void which will be either on edge centre of body centre will be nearest to tetrahedral void on nearest body diagonal.

So, tenth of body diagonal = $\left(8\sqrt{3}\right)\sqrt{3} = 24\text{Å}$,

Let octahedral void is on AD and nearest tetrahedral void is on AX from ¼ of total AX from A

So, coordinate of octahedral centre = (0, 0, $4\sqrt{3}$)

Coordinate of tetrahedral centre = $\left(2\sqrt{3}, 2\sqrt{3}, 2\sqrt{3}\right)$

Distance b/w these voids = $\sqrt{\left(2\sqrt{3}\right)^2 + \left(2\sqrt{3}\right)^2 + \left(2\sqrt{3}\right)^2}$ = 6Å

SECTION – C

48. 00009.74

49. 00048.25

50. 00009.26

Mathematics

PART – III

SECTION – A

51. A

Sol. $CV = \frac{\sigma}{x} \Rightarrow V_1 = \frac{\sigma}{30} \quad V_2 = \frac{\sigma}{25}$

$6V_1 - 5V_2 = \frac{\sigma}{5} - \frac{\sigma}{5} = 0$

52. B

Sol. $Tn = \frac{2}{n(n+1)(n+2)}$

$Tn = \frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}$

On solving

$= \frac{n(n+3)}{2(n+1)(n+2)} \Rightarrow K = 2$

53. A

Sol. $\frac{18!}{6!6!6!}$

54. D

55. C

Sol. $f(x) = \begin{cases} 0 & x \in \mathbb{Z} \\ 1 & x \phi \mathbb{Z} \end{cases}$

$gof = \sin^{-1} 0 = 0$

$\sin^{-1} 1 = \frac{\pi}{2}$

56. D

57. A

Sol. $f(x) = [x] + 2 \Rightarrow f(g(x)) = [\cos x] + 2$

$f\left(g\left(\frac{\pi}{2}\right)\right) = 0$

58. A

59. B

60. C

61. B

Sol. $xy=1$

$$x\frac{dy}{dx}+y=0$$

$$\frac{dy}{dx}=-\frac{y}{x}$$

$$y-y_1=\frac{x_1}{y_1}\left(x-x_1\right)$$

$$yy_1-y_1^2=x_1x-x_1^2$$

$$x_1x-yy_1+y_1^2-x_1^2=0$$

62. D

Sol. $$=\begin{vmatrix}1 & p & q\\ 0 & q-p & r-q\\ 0 & r-q & p-r\end{vmatrix}$$

$$=1\left[\left(q-p\right)\left(p-r\right)-\left(r-q\right)\left(r-q\right)\right]$$

$$=\frac{-1}{2}\left[\left(p-q\right)^2+\left(q-r\right)^2+\left(r-p\right)^2\right]=0$$

$$\Rightarrow p=q=r$$

$$\angle P=\angle Q=\angle R$$

63. B

Sol. $\left(t+1\right)x+8y-4t=0$

$$tx+\left(t+3\right)y-\left(3t-1\right)=0$$

$$\frac{t+1}{t}=\frac{8}{t+3}\pm\frac{-4t}{-\left(3t-1\right)}$$

$$t=3$$

64. A

Sol. $$I\int\frac{3x^2+2x}{\left(x^3+x^2+1\right)^2+2^2}dx$$

$$=\frac{1}{2}\tan^{-1}\left(\frac{x^3+x^2+1}{2}\right)+C$$

65. A

Sol. $$f\left(x\right)=\int_{1/x}^{\sqrt{x}}\sin\left(t^2\right)dt$$

$$f'\left(x\right)=\sin x.\frac{1}{2\sqrt{x}}-\sin\left(\frac{1}{x^2}\right)\left(\frac{-1}{x^2}\right)$$

$$f'(1)=\frac{\sin 1}{2}+\sin 1$$

$$=\frac{3}{2}\sin 1$$

66. C

Sol. $y=\left[\frac{x^2}{36}+1\right]$

$$\text{Area }=\frac{1}{2}(2+3)\times 1$$

$$=\frac{5}{2}$$

67. B

Sol. $(xy-1)\frac{dy}{dx}=-y^2$

$$\frac{xy-1}{y^2}=\frac{-dx}{dy}$$

$$\frac{dx}{dy}=\frac{1}{y^2}-\frac{x}{y}$$

$$\frac{dx}{dy}+\frac{1}{y}x=\frac{1}{y^2}$$

$$I.F.=e^{\int\frac{1}{y}dy}=y$$

$$x.y=\int\frac{1}{y^2}.ydy$$

$$xy=\ln y$$

$$ke=\ln e$$

$$k=\frac{1}{e}$$

68. C

Sol. $n(E)={}^3C_2+{}^3C_2+{}^4C_2+{}^3C_1\times{}^4C_1=24$

$$n(S)={}^{10}C_2=45$$

$$P(E)=\frac{24}{45}\Rightarrow\frac{8}{15}$$

69. B

Sol. $(1+x)^{13}(x+1)^{14}=\left\{{}^{13}C_0+{}^{13}C_1x+{}^{13}C_2x^2+....+{}^{13}C_{13}x^{13}\right\}$

Coeff of x^{13} in $(1+x)^{27}={}^{27}C_{13}={}^{27}C_{14}$

70. B

Sol. $\frac{a+b+c}{3}=A \qquad (abc)^{1/3}=G$

$a+b+c=3A \quad abc=G^3$

$H=\frac{3}{\frac{1}{b}+\frac{1}{b}+\frac{1}{c}} \Rightarrow H=\frac{3abc}{ab+bc+ca} \Rightarrow H=\frac{3G^3}{ab+bc+ca}$

$ab+bc+ca=\frac{3G^3}{H}$

$x^3-3Ax^2+\frac{3G^3}{H}x-G^3=0$

SECTION – B

71. 0

72. 2

Sol. $\begin{vmatrix} 4 & -a & -3 \\ 1 & 2 & a \\ a & 0 & 2 \end{vmatrix}=0$

$4(4-0)+a(2-a^2)+3(0-2a)=0$

$a^3+4a-16=0$

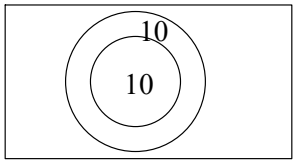
$a=2$

SECTION – C

73. 00018.00

74. 00008.00

Sol. $n(A)=10, n(B)=20$



$n(A\Delta B)=10$

75. 00005.00

Sol. For two common tangents

$|r_1-r_2|<C_1C_2<r_1+r_2$

$r_1=1$

$r_2=\sqrt{16+9-x^2}$

$C_1C_2=\sqrt{16+9}=5$

$\left|\sqrt{25-\lambda^2}-1\right|<5<6\sqrt{25-\lambda^2}+1$

$-3<\lambda<3$